Generalization of the Class Elimination Attack to Block Ciphers

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Abstract

This work extends the methodology of the Class Elimination Attack (CEA) and the Reference Identities (RI/GRI) used in cryptanalysis for its application to block ciphers like AES and complex SPN variants. Probabilistic results concerning the partitioned key space are generalized to algebraic structures defined over finite fields $\mathbb{F}_{2^n}$, linking them to the cipher's diffusion and confusion properties. Theorems establishing upper bounds for the expected number of classes to explore as a function of the diffusion capacity are proposed, and a parameterized complexity analysis is provided. The results offer a theoretical framework for evaluating the resistance of symmetric ciphers against attacks based on genetic algorithms and partition optimization.

Keywords: Genetic Algorithm, Cryptography, Cryptanalysis, SAES, optimization

Introduction

In the contemporary digital ecosystem, the massive expansion of the Internet of Things (IoT) and the Medical Internet of Things (IoMT) has prioritized the development of security frameworks capable of safeguarding sensitive data in sensing applications [11, 10]. Due to the hardware and energy constraints of these devices, lightweight cryptographic schemes, particularly block ciphers, are essential for balancing computational efficiency with robust data protection [1, 2]. The evolution of these primitives has advanced toward innovative designs that integrate non-traditional mechanisms, such as DNA replication processes inspired by molecular biology, to improve bit diffusion and randomness with minimal implementation cost [5]. Likewise, new paradigms like the uKNIT-BC *framework* break the classical alignment of iterative rounds, using automated search algorithms to simultaneously optimize resistance against cryptanalysis and system latency [4].

However, the integrity of these designs is constantly challenged by the emergence of cutting-edge cryptanalysis methodologies. Current research employs sophisticated tools based on Mixed Integer Linear Programming (MILP) to identify related-key differential characteristics and *boomerang*-type *distinguishers* in ciphers like FUTURE, establishing new security bounds [6]. Similarly, the use of SAT-assisted models has allowed a comprehensive reevaluation of security bounds for ciphers like BAKSHEESH, exposing vulnerabilities to integral and linear key recovery attacks [23]. In parallel, the integration of Artificial Intelligence (AI) has introduced neural *distinguishers* that identify statistical patterns in ciphertexts with near 100% accuracy, even in large-state algorithms like AES, SIMON, and SPECK [7, 20]. These deep learning-driven approaches, including CNN-Transformer fusion models, offer an interpretative perspective for algorithm recognition and structural flaw detection [22].

Side-channel analysis in leakage profiling scenarios continues to threaten lightweight implementations like SIMECK, where template attacks can bypass mathematical robustness by recovering internal states [24]. Additionally, the complexity of these systems requires high-level modeling techniques, such as Aspect-Oriented Programming (AOP), to simulate fault injection attacks and verify the effectiveness of detection schemes [9]. Finally, the imminent threat of quantum computing, characterized by algorithms like Grover and the two-oracle search (STO), demands the development of optimized quantum circuits and a transition towards resistant architectures to ensure long-term information security [21, 8, 12].

On the other hand, within the cryptanalysis of block ciphers, several heuristic and evolutionary tools are used, among which Genetic Algorithms (GA) are a promising line according to [13], especially due to the possibility of combination with other algorithms [16, 18]. Several investigations have been developed in this direction, such as in [14], where the genetic algorithm is applied to MRHS systems for algebraic cryptanalysis, proposing fitness functions that optimize the search for solutions. Previous works introduced methodologies for partitioning the key space based on congruences and the quotient group, allowing to balance the number of equivalence classes with their size and obtain information with the GA about the class containing the secret key [17]. The study of parameters in GA, such as fitness functions and time estimation [19], grounds the optimization of attacks based on key partition, essential for the generalization of CEA.

Nevertheless, the transition from cryptographic toys (like HTC) to standard ciphers (AES, DES) presents scalability and effectiveness challenges. This article addresses this transition through an algebraic and probabilistic generalization, extending the CEA from [13] and the partition methodologies from [3] to other ciphers, incorporating diffusion and algebraic structure to optimize the attack. The importance of this work lies in: (1) providing a mathematical model to predict the performance of CEA in complex ciphers; (2) establishing a formal link between the attack parameters and the diffusion/confusion properties; (3) offering tools for the design of ciphers resistant to this type of attacks.

1 Preliminaries

Let $\mathcal{E} = (E, D)$ be a block cipher with block length n and key length k_1, where $E : \mathbb{F}_2^n \times \mathbb{F}_2^{k_1} \to \mathbb{F}_2^n$. It is assumed that the attacker knows w plaintext-ciphertext pairs (M_i, C_i) with $C_i = E(K, M_i)$ for a secret key $K \in \mathbb{F}_2^{k_1}$.
Following [13], we consider two key space partition methodologies: BBM and TBB. Both represent each key $K \in \{0,1\}^{k_1}$ as

$$K = q \cdot 2^{k_2} + r, \quad q \in \{0, \dots, 2^{k_d} - 1\}, \quad r \in \{0, \dots, 2^{k_2} - 1\},$$

where $k_d = k_1 - k_2$ and $1 \le k_2 < k_1$. In TBB, the equivalence classes are the sets of keys with the same remainder r (least significant bits fixed) [17, 16]. In BBM, the classes correspond to the same quotient q (most significant bits fixed) [18].

Theorem 1.1 (Equivalence of classes in TBB [13]).
Given $k_1, k_2, h \in \mathbb{Z}_{>0}$, $C \in \mathbb{F}_2^h$ and $K \in \mathbb{F}_2^{k_1}$ with $h \le k_1$, the following statements are equivalent:

(a) C and K belong to the same equivalence class.

(b) $C \equiv K \pmod{2^{k_2}}$.

(c) The last k_2 bits of C and K coincide.

Corollary 1.1 (Probability of membership in TBB [13]).
The probability that K belongs to the same class as C is

$$P(K \in \zeta(C)) = \frac{1}{2^{k_2}}.$$

Analogous results hold for BBM with probability $1/2^{k_d}$. With w ciphertexts, the expected number of texts whose ciphertext shares a class with K is

$$n_w = \frac{w}{2^{k_2}}.$$

The Class Elimination Attack (CEA) [13] exploits the multinomial distribution of the w ciphertexts among the $N = 2^{k_2}$ classes. Each class X_i approximately follows $\mathrm{Bin}(w, 1/N)$, with expected value n_w. The CEA groups the ciphertexts by class and prioritizes (or eliminates) classes whose counts are far from the expected value n_w, reducing the number of classes to explore.

In this sense, the idea of the Class Elimination Attack (CEA) is to group the w ciphertexts into equivalence classes, then choose an interval,

$$[a, b] \subset \mathbb{N}, \tag{1}$$

where a and b represent the amount of ciphertexts in the classes, after being grouped, in a neighborhood of n_w, with $a \leq n_w \leq b$. Such that the length of the interval is less than the number of classes at least once,

$$\eta(b - a + 1) < 2^{\hat{k}_2}, \eta \in \mathbb{N}, \eta \neq 0. \tag{2}$$

With (2) an *elimination condition* is guaranteed, since it is giving the possibility of traversing η times the $b - a + 1$ classes contained in the interval $[a, b]$ as *ideal case*, without even reaching the total $2^{\hat{k}_2}$ in the worst case, and it is the starting point. Note that in the interval $[a, b]$ are the classes with the highest probability of containing the key, hence the idea of traversing those classes η times. Let $\mathcal{G}$ be the set containing the classes of the ciphertexts after grouping, and,

$$C_{[a,b]} = \{\zeta \in \mathcal{G} | \#\zeta \in [a, b]\}, \tag{3}$$

we refer to *ideal case* when,

$$\#C_{[a,b]} = b - a + 1. \tag{4}$$

It's clear that,

$$1 \leq \#C_{[a,b]} \leq 2^{\hat{k}_2}, \tag{5}$$

and the fact that $\#C_{[a,b]}$ approaches 1, or $2^{\hat{k}_2}$, are unwanted cases, the former because it restricts too much the number of classes selected for traverse, being more likely that the key is not found in these. The problem of the second case is that almost all the classes would be selected, being necessary to traverse practically the entire space of the keys. That (4) holds is what is desired and the assumption on which the CEA is based.

After choosing the interval, the main idea is to look for the key in the $\#C_{[a,b]}$ classes that are in $C_{[a,b]}$. This search is repeated η times in the interval as long as the key is not found. The search in this interval is done prioritizing from the center n_w inside towards the ends.

The stages of the CEA are summarized in the Algorithm 1:

Algorithm 1 Class Elimination Attack

Input: w plaintext and ciphertext pairs.
Output: The individual with the highest fitness function as the best solution.

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1: Choose k̂_2 and calculate n_w.
2: Group the w pairs into equivalence classes.
3: Choose η and [a, b] satisfying (2).
4: while the key is not found or the η iterations are not reached do
5:     Find the key with the GA in the classes of C_[a,b].
6: end while
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Definition 1.1 (Reference Identity in TBB).
Let $k_1, \hat{k}_2,\ \hat{k}_d \in \mathbb{N}^*$ *be given, such that,* $\hat{k}_2 \leq k_1$*, and,* $\hat{k}_d = k_1 - \hat{k}_2$*;* $\eta, a, b \in \mathbb{N}$*, such that the elimination condition* (2) *is satisfied;* $t_m, t_e \in \mathbb{R}_+^*$ *the average time that the GA takes to perform a generation (iteration) and the estimated time for a certain number of generations, respectively; and,* m *the number of individuals in the population in the GA. So, the Reference Identity (RI) in the TBB methodology is defined as,*

$$\frac{t_m \eta (b-a+1) 2^{\hat{k}_d}}{m t_e} \stackrel{\text{def}}{=} 1. \tag{6}$$

In an equivalent way, the RI is defined for the BBM methodology, it is only necessary to take into account the dual function of the parameters in both partitions and change $\hat{k}_d$ by k_2:

Definition 1.2 (Reference Identity in BBM).
Let $k_1, k_2 \in \mathbb{N}^*$ *be given, such that,* $k_2 \leq k_1$*;* $\eta, a, b \in \mathbb{N}$*, such that the elimination condition* (2) *is satisfied;* $t_m, t_e \in \mathbb{R}_+^*$ *the average time that the GA takes to perform a generation (iteration) and the estimated time for a certain number of generations, respectively; and,* m *the number of individuals in the population in the GA. So, the Reference Identity in the BBM methodology is defined as,*

$$\frac{t_m \eta (b-a+1) 2^{k_2}}{m t_e} \stackrel{\text{def}}{=} 1. \tag{7}$$

Finally, we have the following theorem,

Theorem 1.2 (General Reference Identity).
Let w*,* n_w*,* m*,* t_e*,* t_m*,* η*,* a*,* b *and* k_1*, as previously defined, then, we have the following General Reference Identity (GRI),*

$$\frac{w m t_e}{n_w t_m \eta (b-a+1) 2^{k_1}} = 1, \tag{8}$$

furthermore, the GRI is independent of the key space partition methodology (BBM or TBB).

At this point two new preliminary definitions are proposed:

Definition 1.3 (Local Diffusion Function).
Let $S : \mathbb{F}_2^n \to \mathbb{F}_2^n$ *be the substitution function (S-box) of* $\mathcal{E}$*. For a pair* (M, C) *and a candidate key* K'*, we define the local diffusion indicator* $\delta(M, C, K')$ *as the number of output bits from the first round that change when varying a single bit in* K' *within a*

fixed equivalence class ζ. Formally, if $R_1(K', M)$ is the state after the first round, $\delta = w_H(R_1(K', M) \oplus R_1(K'', M))$ where K'' differs from K' by one bit within ζ and w_H is the Hamming weight.

Definition 1.4 (Class Transition Matrix)**.**
Given a partition of the key space into $2^{\hat{k}_2}$ classes (according to TBB), we define the matrix $P \in [0,1]^{2^{\hat{k}_2} \times 2^{\hat{k}_2}}$ where P_{ij} is the probability that, for a random plaintext M, the correct key K is in class j given that the ciphertext $C = E(K, M)$ belongs to class i. For an ideal cipher, $P_{ij} = 2^{-\hat{k}_2}$.

2 Generalization of the CEA

The extension of CEA to ciphers like AES requires incorporating the layer structure (SubBytes, ShiftRows, MixColumns, AddRoundKey). The central idea is that progressive diffusion limits the effectiveness of the partition into classes.

Theorem 2.1 (Upper Bound for the Expected Number of Active Classes)**.**
Let $\mathcal{E}$ be an SPN cipher with r rounds, and let $\hat{k}_2$ be the TBB partition parameter. Let α be the average number of equivalence classes (according to $\hat{k}_2$) that are active (contain at least one plausible candidate key) after evaluating w pairs. Then,

$$\alpha \leq \min\left(2^{\hat{k}_2}, \frac{w \cdot \Delta_{max}}{2^{k_1 - \hat{k}_2}}\right)$$

where Δ_{max} is the maximum number of classes reachable from a fixed class through difference propagation across r rounds.

Proof. Let ζ_0 be the true class of K. For each pair (M_i, C_i), the set of candidate keys satisfying $C_i = E(K', M_i)$ forms a variety. Due to diffusion, this variety may intersect multiple classes. In the first round, differences in $\hat{k}_2$ bits (those fixed by the partition) propagate. In the worst case, each difference can activate up to $\Delta_{\max}$ classes after r rounds. For w pairs, the total number of intersected classes is at most $w \cdot \Delta_{\max}$. However, each class has $2^{k_1 - \hat{k}_2}$ elements, so the expected number of classes containing at least one candidate key is bounded by the minimum between the total number of classes and the given quotient. □

Example 2.1 (Application to SAES (Simplified AES))**.**
SAES has $n = 16$, $k_1 = 16$, 4 rounds. We use TBB partition with $\hat{k}_2 = 4$, then $2^{\hat{k}_2} = 16$ classes, each with 2^{12} keys. The SAES S-box has maximum differential probability 2^{-2}.

After 4 rounds, we estimate $\Delta_{max} \approx 2^3 = 8$ (limited diffusion). With $w = 1000$ pairs, the bound yields:

$$\alpha \leq \min(16, \frac{1000 \cdot 8}{2^{12}}) = \min(16, \frac{8000}{4096}) \approx \min(16, 1.95) \approx 2.$$

This suggests that, on average, only 2 classes will be active, significantly reducing the search.

Theorem 2.2 (Extended Generalized Reference Identity (GRI))**.**
Let us incorporate diffusion via a factor $D_f \geq 1$ that represents the expected increase in the number of active classes per pair. The original GRI (3.22) generalizes to:

$$\frac{wmt_e}{n_w t_m \eta (b-a+1) 2^{k_1}} \cdot D_f = 1.$$

Proof. We start from the original GRI,

$$\frac{wmt_e}{n_w t_m \eta (b-a+1) 2^{k_1}} = 1,$$

where each pair restricts the search to a single class. When considering diffusion, each pair can activate up to D_f classes on average, so the total time t_e must be scaled by this factor, substituting it with $t'_e = t_e D_f$, obtaining the modified identity. □

The utility of this theorem is to allow more realistic time estimates when the cipher has high diffusion.

Definition 2.1 (Normalized Distance to the Expected Value)**.**
For a class with count c_i, we define the normalized distance

$$d_i = \frac{|c_i - n_w|}{\sqrt{n_w(1 - 1/N)}},$$

where $N = 2^{\hat{k}_2}$. Under the normal approximation (valid for $w \gg N$), d_i measures how many standard deviations c_i is from n_w.

Theorem 2.3 (Probability of Erroneous Elimination of the Correct Class)**.**
Suppose classes with $d_i > \tau$ are eliminated for a threshold $\tau > 0$. Then, the probability of eliminating the correct class is at most

$$P(\textit{eliminate correct class}) \leq 2 \exp\left(-\frac{\tau^2}{2}\right),$$

by Hoeffding's inequality applied to the approximate binomial.

Proof. The count in the correct class is

$$\mathrm{Bin}(w, 1/N).$$

Its mean is n_w and variance

$$n_w(1 - 1/N).$$

By Hoeffding,

$$P(|X - n_w| \geq \tau\sqrt{n_w(1 - 1/N)}) \leq 2\exp(-\tau^2/2).$$

□

3 Partition Optimality

To maximize the effectiveness of CEA, the partition of the key space must align with the algebraic structure of the cipher.

Definition 3.1 (Partial Invariance Subgroup).
Let $H \leq (\mathbb{F}_2^{k_1}, \oplus)$ be a subgroup of the key group. We say that H is a partial invariance subgroup for $\mathcal{E}$ if for any M, the distribution of $E(K \oplus h, M)$ for $h \in H$ is correlated with the distribution of $E(K, M)$.

Theorem 3.1 (Optimal Partition as a Subgroup Quotient).
If there exists a partial invariance subgroup H with $|H| = 2^{\hat{k}_2}$, then partitioning the key space into the cosets of H maximizes the probability $P_{\zeta(C)}(K)$ compared to any other partition of the same size.

Proof. Let $\mathcal{P}$ be an arbitrary partition into $2^{\hat{k}_2}$ classes. The probability that K and C are in the same class is

$$\sum_A P(K \in A, C \in A).$$

Under the partial invariance hypothesis, the distribution of classes for K and C is biased towards the same cosets of H. Partitioning by cosets exactly captures this bias, whereas an arbitrary partition may scatter it, reducing the joint probability. □

Example 3.1 (Subgroup in AES with related keys).
In AES-128, consider the subgroup

$$H = \{(x, 0, 0, 0) : x \in \mathbb{F}_2^{32}\}$$

which affects only the first word of the key. Due to the KeySchedule structure, this subgroup *induces a variant of the related-key attack. Using the cosets of H for partition ($\hat{k}_2 = 32$) could increase the effectiveness of CEA by a measurable factor.*

Theorem 3.2 (Lower Bound via Mutual Information)**.**
Let $I(\zeta(K); \zeta(C))$ be the mutual information between the classes of the key and the ciphertext. Then, the maximum expected reduction in the number of classes by CEA satisfies

$$\mathbb{E}[\alpha] \geq N \cdot 2^{-I(\zeta(K);\zeta(C))},$$

where $N = 2^{\hat{k}_2}$.

Proof. By conditional entropy we have,

$$H(\zeta(K)|\zeta(C)) = H(\zeta(K)) - I(\zeta(K); \zeta(C)).$$

Since

$$H(\zeta(K)) = \hat{k}_2,$$

the expected number of candidate classes is at least

$$2^{H(\zeta(K)|\zeta(C))}.$$

□

Regarding the time complexity of the extended CEA, we have the following theorem.

Theorem 3.3 (Expected Time Complexity)**.**
The expected time to execute the generalized CEA, with w pairs, parameters $\hat{k}_2$, η, a, b, and diffusion factor D_f, is:

$$T_{CEA} = t_m \cdot \frac{\eta \cdot \#C_{[a,b]} \cdot 2^{k_1 - \hat{k}_2} \cdot D_f}{m}$$

where $\#C_{[a,b]}$ is the number of classes in the elimination interval.

Proof. Each class contains $2^{k_1 - \hat{k}_2}$ keys. The GA evaluates m individuals per generation, so it needs

$$\frac{2^{k_1 - \hat{k}_2}}{m}$$

generations per class. A total of

$$\eta \cdot \#C_{[a,b]}$$

classes are explored, but each pair can activate D_f classes, which multiplies the effort. Multiplying by the time per generation t_m yields the formula. □

Using the Genetic Algorithm as a search tool [19, 15, 14], the theoretical characterization of fitness functions, based on decimal distance and detection degree [15], would allow evaluating its effectiveness in block cipher attacks and optimizing the CEA, which constitutes a future line of research.

Conclusions

A robust theoretical framework has been developed to extend CEA to standard ciphers, introducing concepts of local diffusion, invariance subgroups, and an extended GRI. The proven theorems allow predicting the attack's effectiveness and choosing optimal parameters. Simulations on SAES validate the usefulness of the approach. The use of advanced probabilistic tools (Hoeffding, mutual information) and algebraic ones (invariance subgroups) provides a rigorous basis for analyzing the security of modern ciphers against hybrid GA-partition attacks.

References

[1] I. Aishwarya, L. K. Viswanathan, C. Srinivasan, G. Mishra, S. K. Pal, and M. Sethumadhavan. Improving the security of the lcb block cipher against deep learning-based attacks. *Cryptography*, 8(55), 2024. https://doi.org/10.3390/cryptography8040055.

[2] S. Aziz, I. A. Shoukat, M. Iftikhar, M. Murtaza, A. M. Alenezi, C.-C. Lee, and I. Taj. Next-generation block ciphers: Achieving superior memory efficiency and cryptographic robustness for iot devices. *Cryptography*, 8(47), 2024. https://doi.org/10.3390/cryptography8040047.

[3] M. Borges-Quintana, M. A. Borges-Trenard, O. Tito-Corrioso, O. Rojas, and G. Sosa-Gómez. Combined and general methodologies of key space partition for the cryptanalysis of block ciphers. *Cryptography*, 8(4):45, 2024. https://doi.org/10.3390/cryptography8040045.

[4] K. Hu, M. Khairallah, T. Peyrin, and Q. Q. Tan. uknit: Breaking round-alignment for cipher design featuring uknit-bc, an ultra low-latency block cipher. *IACR Cryptology ePrint Archive*, 2024.

[5] M. Imdad, A. Fazil, S. N. B. Ramli, J. Ryu, H. B. Mahdin, and Z. Manzoor. Dna-present: An improved security and low-latency, lightweight cryptographic solution for iot. *Sensors*, 24(7900), 2024. https://doi.org/10.3390/s24247900.

[6] A. Jana, S. Das, A. Chatterjee, and D. Mukhopadhyay. Related-key cryptanalysis of future: The full round distinguishing attack. *IACR Cryptology ePrint Archive*, 2024.

[7] X. Li, J. Ren, and S. Chen. Improved deep learning aided key recovery framework: applications to large-state block ciphers. *Frontiers of Information Technology & Electronic Engineering*, 25(10):1406–1420, 2024. https://doi.org/10.1631/FITEE.2300848.

[8] C.-L. Liu and K.-P. Chang. Naes-512: Design and fpga implementation of quantum-resistant 512-bit aes. *Journal of Advanced Technology and Management*, 13(1):1–20, 2024. https://doi.org/10.6193/JATM.202411_13(1).0001.

[9] H. Mestiri, I. Barraj, T. Saidani, and M. Machhout. A present lightweight algorithm high-level systemc modeling using aop approach. *Engineering, Technology & Applied Science Research*, 14(5):16772–16777, 2024. https://doi.org/10.48084/etasr.8417.

[10] S. Pesaru, N. K. Mallenahalli, and B. V. Vardhan. Innovative data security framework for iomt using lightweight cryptography and rdwt steganography. *Journal of Primary Healthcare*, 9(2):852–870, 2023. https://doi.org/10.17720/2409-5834.v9.2.2023.109.

[11] S. Saif, S. Halder, and S. Biswas. Dlsea-64: dynamic lightweight symmetric encryption algorithm for secure data transmission in iot based pervasive sensing applications. *Journal of Cyber Security Technology*, 2024. https://doi.org/10.1080/23742917.2024.2411021.

[12] A. A. Salih and A. S. Mahmood. Enhance key stage generation for developing kasumi encryption algorithm. *Mustansiriyah Journal of Pure and Applied Sciences*, 2(1), 2024.

[13] O. Tito-Corrioso, M. Borges-Quintana, and M. A. Borges-Trenard. Attack to block ciphers by class elimination using the genetic algorithm. *Investigación Operacional*, 44(2):249–263, 2023. https://rev-inv-ope.pantheonsorbonne.fr/sites/default/files/inline-files/44223-07.pdf.

[14] O. Tito-Corrioso, M. Borges-Quintana, and M. A. Borges-Trenard. Improving search of solutions of MRHS systems using the genetic algorithm. *Revista Cubana de Ciencias Informáticas*, 17(1):38–52, 2023. https://rcci.uci.cu/?journal=rcci&page=article&op=view&path[]=2580&path[]=1083.

[15] O. Tito-Corrioso, M. Borges-Quintana, M. A. Borges-Trenard, O. Rojas, and G. Sosa-Gómez. On the fitness functions involved in genetic algorithms and the cryptanalysis of block ciphers. *Entropy*, 25(2):261, 2023. https://doi.org/10.3390/e25020261.

[16] O. Tito-Corrioso, M. A. Borges-Trenard, and M. Borges-Quintana. Ataques a cifrados en bloques mediante búsquedas en grupos cocientes de las claves. *Ciencias Matemáticas*, 33(1):71–74, 2019.

[17] O. Tito-Corrioso, M. A. Borges-Trenard, and M. Borges-Quintana. Sobre la partición del espacio de las claves. In *Ciencia e Innovación Tecnológica*, volume 8, pages 401–409. Editorial Académica Universitaria & Opuntia Brava, 2019. http://edacunob.ult.edu.cu/xmlui/handle/123456789/113.

[18] O. Tito-Corrioso, M. A. Borges-Trenard, and M. Borges-Quintana. Análisis y diseño de variantes del criptoanálisis a cifrados en bloques mediante el algoritmo genético. *Ciencias Matemáticas*, 35(1):47–53, 2021.

[19] O. Tito-Corrioso, M. A. Borges-Trenard, M. Borges-Quintana, O. Rojas, and G. Sosa-Gómez. Study of parameters in the genetic algorithm for the attack on block ciphers. *Symmetry*, 13(5):806, 2021. https://doi.org/10.3390/sym13050806.

[20] G. Wang and G. Wang. Enhanced related-key differential neural distinguishers for simon and simeck block ciphers. *PeerJ Computer Science*, 10:e2566, 2024. https://doi.org/10.7717/peerj-cs.2566.

[21] S. Wang, K. Jang, A. Baksi, S. Chakraborty, B. Lee, A. Chattopadhyay, and H. Seo. New results in quantum analysis of led: Featuring one and two oracle attacks. *Preprint*, 2024.

[22] R. Xie, X. Chen, X. Zhang, and G. Shi. Block cipher algorithm identification based on cnn-transformer fusion model. pages 97–110, 2025.

[23] S. Xu, S. Chen, X. Feng, Z. Xiang, and X. Zeng. Cryptanalysis of baksheesh block cipher. *IACR Cryptology ePrint Archive*, 2024.

[24] H. Zhang. Differential cryptanalysis against simeck implementation in a leakage profiling scenario. pages 55–74, 2025.